

Rethinking Biological Weapon Deterrence in the 21st Century

Daniel Allen
dallen170201@gmail.com

Abstract

Biological weapons (BW)s have traditionally been considered of limited strategic utility; they are uncontrollable, indiscriminate, and unpredictable, and their effects – even when successful – are often more delayed than other weapons systems. Advances in AI have sparked a new generation of possible biological weapons, ones that might free themselves from the strategic shackles of the past. A host of actors from non-state to great powers will find utility for these weapons, and with an international regime struggling to control the proliferation of knowledge, technology, and capability, we must revisit the discussion of how to deter the development and use of BWs in the 21st century.

Introduction

Historically, military strategists have judged biological weapons to have limited utility. Most pathogens incubate for days to weeks, far behind modern operational tempos; environmental factors such as wind, temperature, and solar radiation erode the reliability of dispersion; and contagious agents risk back-infection ("blowback") on friendly forces or general populations. Their production is highly specialized, with 20th-century BW programs requiring lavish funding (the Soviet program was estimated to have spent 35 billion dollars between 1972 and 1992)¹, thousands of man-hours in the lab performing finicky strain manipulation, and whole-state enterprises. Even then, programs were not always successful.² As summarized succinctly by Pfizer's John Smith on the mercurial character of living organisms: "The mold is as temperamental as an opera singer, the yields are low, the isolation is difficult, the extraction is murder, the purification invites disaster, and the assay is unsatisfactory."³ Field trials at U.S. Army Dugway Proving Ground showed simulant agent clouds drifting unpredictably for tens of kilometers while tabletop exercises such as "Dark Winter" demonstrated that once symptoms appear in human victims, cascading social and economic disruption could swamp any tactical benefit.⁴ As such, planners view BWs primarily as instruments of terror or covert sabotage rather than decisive battlefield weapons.^{5, 6}

Today, the rapid evolution of artificial intelligence (AI) threatens to erode these barriers of poor targetability and limited persistence, offering a host of possible new attack vectors that state and non-state actors (NSAs) could exploit for tactical and strategic effect. As these weapons become more capable and more accessible, they become more difficult to deter.

¹ Barriers to Bioweapons, p.35

² See the limited success of the American BW program during the Cold War

³ Barriers to Bioweapons, Chapter 1, p.6

⁴ <https://centerforhealthsecurity.org/our-work/tabletop-exercises/dark-winter-a-training-tabletop-exercise>

⁵

<https://www.marines.mil/Portals/1/Publications/MCWP%203-37%20MAGTF%20Nuclear%2C%20Biologic%20and%20Chemical%20Defense%20Operations.pdf>

⁶ <https://pmc.ncbi.nlm.nih.gov/articles/PMC7123542/>

Attribution and resilience⁷ become harder when synthetic genomes bear no natural lineage; strategic stability erodes as rival powers fear deniable, high-impact strikes; and the BWC's lack of mandatory verification measures exacerbates uncertainty.

In short, policy has lagged behind technology. Current dual-use and gain-of-function⁸ guidelines were crafted for 20th-century biology and struggle to address AI-designed pathogens or on-demand gene printing^[OB]. Together, these trends fracture the old (though at times spurious) correlation between state capacity to field BW programs, and its BW capability.

Purpose of This Paper

I argue that in this emerging threat environment, we have yet to establish effective heuristics in imagining strategies of attack and defense. While biological weapons are fundamentally different from either chemical or nuclear weapons (and are at best complementary to broader strategies), BWs are a fundamental part of the WMD arsenal. Due to this categorization, I believe it is useful to frame BW use within a deterrence framework (capability, commitment, communication) to help us understand how to deter and de-escalate crises where BWs might be employed.

This paper seeks to answer the question of the extent to which emerging AI-enhanced biological weapons disrupt traditional deterrence logic.

I focus in particular on nuclear deterrence (though do not exclude conventional deterrence) due to its prevalence in deterrence studies, as well as its historical connotations as a deterrent for WMDs. Furthermore, as I argue throughout the paper, unacceptable punishment for an actor seeking to use BWs likely narrows down to massive conventional retaliation, or a nuclear strike.

This paper proceeds in five steps. First, it outlines the key concept of deterrence and the concept's evolution. Second, it situates biological weapons within the broader WMD framework and examines why WMDs are uniquely entwined in “deterrence logic”, and the relation between deterrence and BWs specifically. Third, drawing on Schelling's theory of deterrence — built on capabilities, commitment, and communication — the paper shows how this logic falters when confronted with AI-enabled biological weapons. Fourth, I explore their potential use in grey-zone warfare, where limited applications below the traditional threshold can still generate catastrophic escalation. Finally, it advances eleven mitigation strategies for the short, medium, and long-term to address weaknesses in our ability to deter BWs.

⁷ I.e. the creation of biological countermeasures like vaccines, and the creation of effective non-pharmaceutical interventions (NPIs) like quarantine rules.

⁸ Gain of function research refers to research that optimizes for a particular trait of a pathogen (virulence, for example).

What is Deterrence?

Deterrence is an exercise in engineering belief. In classical terms, it is a strategy aimed at preventing an adversary from taking an action by instilling doubt or fear of the consequences. Thomas Schelling famously described deterrence as a form of coercive bargaining⁹: the art of convincing an opponent not to do something by credibly threatening punishment if they do.¹⁰ Practically, effective deterrence rests on three interlocking requirements: (1) capability — the material ability to impose costs or deny benefits; (2) credible commitment — the demonstrated resolve that the capability will be used if a red line is crossed; and (3) communication — the transmission of that conditional threat to the specific audience whose behavior you seek to shape.¹¹ This triptych is most closely associated with Thomas Schelling's work, who emphasized that deterrence is about shaping expectations *ex ante*, often by creating risk that escalation could slip partly “out of control”, or in his words, the “threat that leaves something to chance”.

Early deterrence theory emerged in the nuclear context, and it is for this reason that we often perceive the two as inextricably linked. Bernard Brodie, the pioneer of nuclear deterrence, laid out the connection between the awesome power of the atom and the inability to use it, writing, “Thus far the chief purpose of our military establishment has been to win wars. From now on, its chief purpose must be to avert them. It can have almost no other useful purpose.” Other military theorists built upon, and at times challenged, Brodie's convention. Herman Kahn, another pioneer, developed “escalation ladders” to examine how a nuclear war might be fought and limited. Kahn's reasoning was that making deterrent threats credible sometimes required preparing to fight if deterrence broke down. He criticized any facile assumption that nuclear threats would automatically be believed; instead, Kahn argued, a state must demonstrate both the capability and willingness to actually use force, even nuclear force, for deterrence to hold.

Although the public tends to associate deterrence with nuclear weapons and superpower crises, the concept long predates the atomic era and applies across many domains. Deterrence can target interstate invasion, cyber operations, or sabotage; its currency need not be nuclear. What matters is whether the defender can make the aggressor believe that crossing a specified threshold will reliably trigger a response that makes the attempt net-negative. RAND's synthesis captures the modern view: success typically requires clarity about the forbidden act, visible means to impose costs or negate benefits, and steps that convince the would-be aggressor the response will actually occur.¹²

⁹ <https://www.nti.org/analysis/articles/disincentivizing-bioweapons-theory-and-policy-approaches/>

¹⁰

https://www.rand.org/content/dam/rand/pubs/perspectives/PE200/PE295/RAND_PE295.pdf#:~:text=Deterrence%20by%20punishment%2C%20on%20the.Steps%20taken%20to%20deny%2C%20such

¹¹

https://nipp.org/information_series/david-j-lonsdale-extended-deterrence-back-to-the-future-no-541-december-1-2022/#:~:text=is%20a%20process%20that%20comprises.better%20understand%20the%20challenges%20of

¹² https://www.rand.org/content/dam/rand/pubs/perspectives/PE200/PE295/RAND_PE295.pdf

Two conceptual cautions follow. First, deterrence success is inherently hard to “prove” because the counterfactual – what the adversary would have done absent the threat – is unobservable.¹³ This epistemic problem is worse when the space of possible unwanted actions is large and partially covert (such as sabotage, subversion, or clandestine use of force). Policymakers must therefore contend with uncertainty, rarely getting clear feedback that their deterrence posture is effective, except in the breach, when deterrence fails. This puts a premium on prudence and worst-case analysis: assume deterrence might fail, and prepare accordingly (a logic that again influenced Herman Kahn and others to advocate robust defense and response options, not just reliance on threat-making). Secondly, misperception can undo deterrence. Robert Jervis argued that leaders often incorrectly toggle between a deterrence model (believing firmness deters) and a spiral model (believing firmness provokes). In practice, states misread signals, misestimate risk tolerances, and sometimes escalate or appease at the wrong times.¹⁴ The point is not to say that deterrence is immeasurable or unusable, rather, that deterrence is not a static property of force structure but a dynamic relationship between minds under uncertainty.

Because the international system is anarchic, there is no higher authority to guarantee promises or punish violations. Thus deterrence becomes a primary tool for ordering expectations around the most consequential acts — the ones that could upend regional balances or destabilize the global system. Historically, nuclear weapons epitomized this category because of their speed, scale, and undeniability. In today’s multipolar world of emerging threats, the concept of deterrence is again evolving to address new WMD threats (like bioweapons or radiological weapons) and even non-physical threats (like strategic cyberattacks) that could wreak global havoc. Therefore, the category of what we deter is, in truth, much broader; any capability that can cause outsized, hard-to-reverse harm, especially with ambiguous authorship, invites the logic of deterrence and the need to choreograph capability, commitment, and communication with unusual care.

Biological Weapons as WMDs

The term weapon of mass destruction (WMD) is frequently employed sloppily, being used by the U.S. government to describe everything from nuclear weapons to pipe bombs.¹⁵ For this paper, I follow the usage anchored in international treaty practice and U.S. strategy documents. WMD comprises nuclear, chemical, and biological weapons (sometimes extended to radiological), not because of a casualty threshold in any given use, but because of the type of weapon and its uniquely destabilizing effects on international order and constraints codified in law.

¹³ https://www.rand.org/content/dam/rand/pubs/research_reports/RR2400/RR2451/RAND_RR2451.pdf, p.17

¹⁴ <https://www.degruyterbrill.com/document/doi/10.1515/9781400885114-006/>

¹⁵ NDU Paper Carus, p.30

Biological weapons are unequivocally categorized as weapons of mass destruction (WMDs), even though they occupy a unique place in the WMD spectrum. Unlike nuclear detonations, which almost invariably cause massive instantaneous destruction, biological weapons might kill many or relatively few, depending on how they are used. At one extreme, a state or group could attempt a mass-casualty strike: for instance, aerosolizing a highly lethal pathogen over a dense city. Several analyses in the 1990s warned that 50 kg of anthrax spores released under the right conditions could inflict casualties on the order of a one-megaton nuclear bomb (1 - 3 million people).¹⁶ Similarly, a small quantity of a potent biotoxin can produce mass effects. For example, 5 kg of botulinum toxin introduced into a municipal water supply (perhaps by contaminating a reservoir) could poison millions of liters of drinking water; as one analysis noted, just half a kilogram of certain agents can render an enormous volume of water deadly.¹⁷ These scenarios, while theoretical, drive home the point that BWs enable an attacker to generate huge casualties with minimal material input, thus making them attractive to opportunistic proliferators. Kalpana Chittaranjan, writing in *Strategic Analysis*, emphasizes this disproportionate potential of BWs, which helps explain why some rogue regimes and non-state actors have pursued them as a shortcut to strategic leverage, or as she references, the “poor man’s nuke”¹⁸.

But biological weapons are not exclusively about creating mass casualties, and can be tailored for destabilizing purposes short of apocalyptic plagues. For instance, BWs can be used for economic warfare by targeting agriculture. Pathogens that infect crops or livestock might not kill a single human directly, yet qualify as WMDs in how they could devastate a nation’s food supply and economy.¹⁹ U.S. Senator Joni Ernst recently warned that an adversary could seek to weaponize such agricultural diseases to cause food insecurity.²⁰ During the Cold War, both the United States and the USSR researched anti-crop and anti-livestock agents²¹, and during the Iran-Iraq War, Saddam Hussein developed a variety of fungal agents to attack Iran’s staple food crops.²²

By the same token, “incapacitating” biological weapons form another category in the spectrum. These are agents intended not necessarily to kill but to sicken and disable. *Coxiella burnetii* (Q fever) has long been cited as a candidate because it can be aerosolized and produce large outbreaks of debilitating illness with low acute mortality; reviews put mortality around 1–2% in the absence of complications, with disability (e.g., prolonged

¹⁶ https://ciaotest.cc.columbia.edu/olj/sa/sa_98chk02.html

¹⁷ I. Malek, “Biological Weapons,” in Steven Rose, ed., *CBW: Chemical and Biological Warfare*, (Boston, Beacon Press: 1969), p. 51.

¹⁸ https://ciaotest.cc.columbia.edu/olj/sa/sa_98chk02.html#note1

¹⁹ https://www.rand.org/pubs/research_briefs/RB7565.html

²⁰ Ibid.

²¹ http://ndupress.ndu.edu/Portals/68/Documents/occasional/cswmd/CSWMD_OccasionalPaper-12.pdf, p.25, 34

²²

<https://biosecurity.fas.org/education/dualuse-agriculture/1.-agroterrorism-and-foodsafety/biowarfare-against-agriculture.htm>

fatigue) a salient effect for planners considering operational impact.²³ A state might calculate that incapacitating a city, the [military personnel] of a geographical chokepoint, or critical support for defense industries by making people too ill to resist could achieve military goals without the international opprobrium that mass fatalities would bring. Underscoring the value of such an arsenal, the U.S. developed *Brucella suis* as a BW agent during the 1940s–50s²⁴, and weaponized Venezuelan equine encephalitis virus (VEEV) in the Cold War era as a primarily incapacitating agent that typically causes severe but self-limited febrile illness.²⁵

Perhaps most alarming for the future is the prospect of engineered pathogens that are tailored to specific populations or even individuals, difficult to attribute, and increasingly customizable in their effects. Advances in genomics and gene-editing (CRISPR and related technologies) have opened discussions about “ethnically targeting” or personalized bioweapons. While largely theoretical at this stage, the concept is taken seriously by security experts.²⁶ In 2022, U.S. Congressman Jason Crow warned that “there are now weapons in development that are designed to target specific people”²⁷. Likewise, an ethnic bioweapon might be engineered to exploit genetic traits more prevalent in one population than others, enabling a pathogen that sickens or kills one population while sparing another. History provides a sobering parallel, showing that governments might find use for such capabilities. In apartheid South Africa, “Project Coast” sought to develop biological agents that would harm black individuals, while leaving whites unaffected.²⁸ While Project Coast never developed such capabilities successfully, today the collection of genetic data across populations (for example, via DNA testing companies or hacked medical databases) raises the possibility that adversaries could map out these genetic susceptibilities. A 2021 Naval Institute analysis noted that adversaries are indeed interested in Western genomic data²⁹, and there is concern that DNA data from genealogy services (like 23andMe) could be “collected by our adversaries for the development of these [targeted bio] systems”.²⁹ To be clear, no confirmed use of such a weapon exists to date, and technical hurdles are significant³⁰ – human genetic variation is complex, and traits are not so neatly exclusive. However, the rapid pace of biotechnology is eroding those hurdles, and advances in machine learning might proliferate this advanced know-how to a swath of actors.³¹ While such capabilities may still be years away, their mere possibility contributes to the sense that

²³ <https://pubmed.ncbi.nlm.nih.gov/14592601/>

²⁴

<https://medcoeckapwstorprd01.blob.core.usgovcloudapi.net/pfw-images/dbimages/Bio%20Ch%2007.pdf>

²⁵ https://oeps.wv.gov/bioterrorism/documents/hcp/USAMRIID_BlueBook.pdf, p.69

²⁶ <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2024.1382356/full>

²⁷ South Africa’s Chemical and Biological Weapons, Homeland Security Digital Archive, p.34-35

²⁸ <https://www.usni.org/magazines/proceedings/2021/june/synthetic-bioweapons-are-coming>

²⁹ <https://www.axios.com/2022/07/24/bio-weapons-target-us-joni-ernst-jason-crow>

³⁰ <https://www.rand.org/pubs/perspectives/PEA3886-1.html>

³¹

<https://thebulletin.org/2023/11/cant-quite-develop-that-dangerous-pathogen-ai-may-soon-be-able-to-help/#post-heading>, Countering WMDs, p.30,

biological weapons could become more attractive and “high-tech” in the coming decades. No longer a low-prestige tool of rogue states alone (a previous theory to explain their lack of use and development³²), cutting-edge bio-weapons might confer a perverse kind of status on their wielders, showcasing biotechnological prowess.

Alongside the immediate material impact of such attacks, we must also contend with the psychological impact of BW use. Even a small-scale bioterror attack can induce disproportionate terror and disruption.³³ The 2001 anthrax letter attacks in the U.S., which killed “only” 5 people, nevertheless caused nationwide panic. 30,000 people took prophylactic antibiotics, postal facilities shut down for decontamination, and resulted in nearly \$1 billion³⁴ in cleanup costs and economic spending.³⁵ In that sense, any use of BW – however limited – has mass disruption effects that are strategic in scope. This psychological power is precisely why extremist groups have eyed biological agents. As UK thinktank Chatham House noted, biological weapons are the “ultimate terrorists weapon”³⁶ because they instill fear far beyond their direct physical impact^[66]. The fear factor amplifies the coercive value of BWs, making even a crude attempt a potent tool of terror.

Given the wide spectrum of use cases, it is tempting to reserve the “WMD” label only for the top-end, mass-casualty cases. To do so, however, would precisely ignore the definition of WMDs established by the international community. The analysis of Seth Carus at the WMD Center at the National Defense University stress that sliding to a purely “effects-based” definition — i.e., “WMD-as-any-weapon-that-causes-mass-destruction” — clashes with how states have actually structured nonproliferation regimes.³⁷ By treaty logic, any possession or use of a biological agent as a weapon falls within the WMD category, irrespective of whether a specific attack kills dozens or thousands. This design avoids perverse incentives (e.g., “legally permissible” small BW arsenals) and reflects a prudential judgment: that even “limited” biological attacks can have runaway second-order effects epidemiological, economic, political, and militarily, that make the category uniquely destabilizing.

The scenarios elaborated upon above are by no means exhaustive. Strategists have mapped out countless attack vectors, and the opportunities for different actors with different goals will only grow over time. This growth yields two practical implications for deterrence. First, because BW effects are often stochastic and partially hidden (incubation delays, environmental sensitivity, noisy signals), the defender’s problem is not just preventing a large attack but shaping the attacker’s calculus across many (now-increasing) incremental options. Second, classification as WMD is not merely semantic but ties BWs to a body of law

³² <https://www.nti.org/analysis/articles/disincentivizing-bioweapons-theory-and-policy-approaches/>

³³ https://www.dhs.gov/xlibrary/assets/prep_biological_fact_sheet.pdf

³⁴ <https://nap.nationalacademies.org/read/11324/chapter/6#59>, p.59 (for 1 bil figure)

³⁵ <https://www.cdc.gov/mmwr/PDF/wk/mm5044.pdf>

³⁶

<https://www.chathamhouse.org/sites/default/files/public/Research/International%20Security/cbrn0207.pdf>, p.14

³⁷ Defining weapons of mass destruction paper

and strategy in which states have reserved the right to respond with severe measures to any WMD use, up to and including nuclear responses in extreme circumstances. Whether such responses would be wise or credible in particular cases is debated, but the declared posture is part of the deterrent environment and therefore cannot be discounted.

Historical Models of Deterrence against BWs

History does not present us with a single, polished policy of deterrence against biological weapons. Instead, the deterrence regime occurred piecemeal in the form of reactive, practical adaptations by the United States and allies as they confronted state programs, gray-zone coercion, and sporadic terrorist use. Throughout, many of the most consequential moves were not labeled “deterrence” but nonetheless functioned to raise adversary costs, constrain opportunity, and signal consequences.

Post-Cold War

Prior to Operation Desert Storm in 1991, there had been no coherent policy from the Department of Defense for the deterrence of regional actors with WMDs.³⁸ Deterrence had been understood exclusively vis-à-vis the Soviet Union and its massive arsenals, and though even the USSR had operational plans to use WMDs such as biological weapons below the strategic level, the United States' policy did not explicitly address these. Instead, in a thread that continues today, U.S. policy makers and her allies opted for ambiguous, flexible responses. A 1968 NATO strategy document (MC 14/3) explains that “NATO should retain the initiative to use nuclear weapons under conditions where it is militarily or politically required.”³⁹ Simultaneously, however, the document also stressed that covert action – such as the one the Warsaw Pact planned to use BWs in – ought to be addressed by the individually affected NATO country.⁴⁰ While the use of BW surely preceded larger military operations, the lack of clarity on NATO's part throughout the escalation highlights the desire to meet a broad range of attack thresholds not through precise policy, but through ambiguous language that persists in U.S. doctrine today – reflecting a continued reluctance to draw clear red lines below the strategic threshold.

In the 1990s, the United States was faced with Iraq: a regional actor willing to use and equipped with a WMD arsenal outside of NATO boundaries. In this period it began the first steps towards building explicit responses for biological weapon attacks. Much like during the Cold War, the threat of nuclear use was pivotal in deterring the use of BWs in the short term. Some scholars argue that “Secretary of State James Baker's implicit threat of a nuclear response contributed to Saddam Hussein's decision not to employ [...] biological weapons that he certainly did possess during the first Persian Gulf War.”⁴¹ Still, these policies were improvisational and leaned on the United States' nuclear arsenal; which might be

³⁸ Countering WMD, p.1

³⁹ <https://www.nato.int/docu/stratdoc/eng/a680116a.pdf>, p.363

⁴⁰ Ibid., 364-465

⁴¹ Counter WMD, p.33

considered an overall inelegant, and potentially noncredible, solution considering the vast conventional superiority of the U.S. armed forces, and the number of options available through economic and political means.

Emerging Policy – 2000-2010

The rise of non-state actors' use of biological agents throughout the 80s - 2000s (the 1986 Rajneesh cult attacks, the Aum Shinrikyo cult through 1990-1994, and the 2001 anthrax attacks)⁴², proved to be the catalyst necessary for larger, institutional approaches towards deterring the threat of bioweapons. U.S. policymakers realized the need to protect beyond military forces⁴³ to include civilians and the broader security infrastructure. This period also marked a shift away from deterrence by punishment to a deterrence by denial strategy; likely spurred by the need to protect a wider range of targets critical to any sort of kinetic response (the punishment aspect of deterrence).

In 2001, the PATRIOT ACT §817 criminalized possession of select biological agents and toxins without proper justification, effectively narrowing legitimate access and creating new penalties for unauthorized ownership.⁴⁴ Under the Clinton administration, multiple initiatives began, such as the Department of Homeland Security's BioWatch program, which was launched across more than 20 major U.S. cities "to detect a biological attack as soon as possible after it has occurred so as to maximize the time available for treatment."⁴⁵ Simultaneously, the National Strategic Stockpile (made up of medical countermeasures suitable for responding to WMD attacks) were made more robust. The U.S. Department of Health and Human Services (DHHS) also launched the BioShield program in 2004, intended to "help incentivize private industry to develop vitally needed medical countermeasures by providing multi-year funding to support advanced research, clinical development, manufacture and procurement."⁴⁶ Finally, new laboratories like the National Biodefense Analysis and Countermeasures Center (NBACC) were established to aid in the rapid attribution of biological and chemical attacks, as well as preempt possible pathways towards developing biochem weapons.⁴⁷ In sum, the early 2000s were pivotal for the creation of a national strategy of *defense* against BWs, and with this strategy, the internal bureaucracy necessary to sustain it.

Biosecurity Regimes – 2010-2016

Under Obama, the shift from biodefense to biosecurity continued. The 2009 document National Strategy for Countering Biological Threats maintains three throughlines in its

⁴² <https://www.ncbi.nlm.nih.gov/books/NBK570614>

⁴³ Countering WMD, p.36

⁴⁴ <https://www.congress.gov/107/plaws/publ56/PLAW-107publ56.htm>

⁴⁵ Ibid., 36

⁴⁶ <https://aspr.hhs.gov/AboutASPR/ProgramOffices/BARDA/Pages/Project-Bioshield.aspx>

⁴⁷ Counter WMD, 32

strategy broadly summarized as a focus on 1) prevention, 2) international cooperation, and 3) integration of key agencies. Building on those pillars, the administration recast deterrence against BWs as in need of a systematic solution that was about “countering” rather than “detering”. The 2009 strategy elevated prevention and dual-use governance (not just consequence management), made international cooperation a core method rather than an afterthought (nearly half of its 77 actions required external partners), and cemented microbial forensics and joint criminal–epidemiological investigations as front-row tools to enable prosecution (punishment) and to raise the perceived risk of getting caught (denial via deterrence and faster termination of attacks). Obama’s 2009 policy committed to a national R&D agenda for microbial forensics and designated NBACC as the federal lead for bioforensic analysis in support of law-enforcement. The Obama administration also attempted to integrate communities that didn’t naturally talk. The strategy calls for systematic cooperation among public health, life sciences, intelligence, and law enforcement; for example, the FBI’s WMD Coordinators expanding outreach to community biology groups and local institutions, because early warning often surfaces in scientific or clinical networks before it is known in the intelligence community.

Interestingly, Obama’s administration also moved the onus of deterrence by denial beyond the responsibility of the United States, enlisting the broader international community. Obama re-energized the BWC as the “premier forum” for managing bio-risk, sending the first U.S. Secretary of State to address a Review Conference and pushing transparency, reporting, and capacity-building, while declining to resurrect a legally binding verification protocol the United States judged unworkable for biology. Where Clinton’s late-1990s strategy tried to build a verification regime around inspections and declarations, Obama pursued governance and implementation—norms and reporting as the scalable way to raise costs and reduce options for misuse in a rapidly globalizing life-sciences ecosystem.

Finally, Obama abandoned the earlier approach of deterring a variety of WMDs with nuclear capabilities. In his 2010 Nuclear Posture Review, the Obama administration pledged not to use or threaten nuclear weapons against Non-Proliferation Treaty (NPT)-compliant non-nuclear states even in the face of CBW attack, while retaining a narrow deterrent role for non-compliant or nuclear-armed states—thereby keeping punishment available for egregious state cases but shifting the day-to-day deterrence burden to non-nuclear tools (attribution, conventional strike, defenses, and resilience).

In short, Obama’s posture sat decisively on the denial side of the deterrence ledger, with punishment present but conditional. This sits in contrast with the 1990s emphasis on arms-control verification and with Cold-War practice that implicitly folded BW contingencies under broad nuclear ambiguity.

Technological Challenges and Volatile Policy – 2016-2025

The last decade has seen administrations attempt to integrate deterrence by denial and deterrence by punishment as a continuous spectrum of deterrence options; though policy

has lacked continuity on previously implemented policy. Under the first Trump administration, the publication of the National Biodefense Strategy suggested a continued building upon Obama's deterrence by denial strategy with an emphasis on global cooperation, national preparedness, and the development of medical countermeasures using private industry.⁴⁸ Simultaneously, in the 2018 National Strategy for Countering WMD Terrorism report, the Trump administration hinted at its own skepticism towards a deterrence by denial approach, admitting that "despite our technological and military advantages, we cannot eliminate all pathways for terrorists to conduct a WMD attack against the United States and its interests".⁴⁹ This ambivalence (which might be extrapolated to state actors, too) might help explain the administration's decision to include a range of non-nuclear attacks (implicitly including biological) towards the U.S. and its allies as falling under the shadow of nuclear deterrence in the case "of catastrophic use"; while remaining vague on specific red lines. The 2018 Nuclear Posture Review (NPR) reversed Obama's narrowing of nuclear weapon use and reintroduced strategic ambiguity.⁵⁰

The Biden administration had to contend with a rapidly changing technological landscape, and centered biodefense as a core national security issue. In the administration's 2022 NPR, Biden maintained Trump's ambiguity of nuclear retaliation in the face of strategic non-nuclear attacks, while simultaneously (and somewhat contradictory) attempting to underscore that the primary role of U.S. nuclear weapons was to deter a nuclear attack; a shift away from the previous administration's "hedge against an uncertain future".⁵¹ It reinstated the U.S. commitment to not use or threaten the use of nuclear weapons against non-nuclear, NPT-compliant states.⁵² Simultaneously, it emphasized "a narrow range of contingencies in which U.S. nuclear weapons may still play a role in deterring attacks that have strategic effects".⁵³ What this threshold specifically meant was left ambiguous. Outside of nuclear deterrence, the Biden administration also pushed to expand its arsenal of integrated deterrence by punishment and denial.

In the (first of its kind) 2023 Biodefense Posture Review (BPR), the Biden administration mentions deterrence over 30 times across the 50-page document. The BPR's most cogent definition of its deterrence posture is offered as:

"A Total Force resilient to biothreats and biological hazards (biohazards)[...] to deter the use of bioweapons and deliberate attack. Such resilience, properly messaged and demonstrated, bolsters

⁴⁸

<https://www.rand.org/pubs/commentary/2018/10/achieving-the-trump-administrations-national-biodefense.html>

⁴⁹ <https://www.armscontrol.org/act/2019-01/news/us-sets-strategy-against-wmd-terrorism>

⁵⁰ <https://media.defense.gov/2018/feb/02/2001872877/-1/-1/1/executive-summary.pdf>, p.3-4

⁵¹

<https://media.defense.gov/2022/Oct/27/2003103845/-1/-1/1/2022-NATIONAL-DEFENSE-STRATEGY-NP-R-MDR.pdf>, p.3 (of NPR section)

⁵²

<https://media.defense.gov/2022/Oct/27/2003103845/-1/-1/1/2022-NATIONAL-DEFENSE-STRATEGY-NP-R-MDR.pdf>, p.9 (of NPR section)

⁵³ *Ibid.*, p.8

*integrated deterrence. DoD's extensive biodefense capabilities can be leveraged to deny or greatly minimize the benefit of using bioweapons and further deter the development or proliferation of bioweapons.”*⁵⁴

The BPR is a good starting point, emphasizing denial and, failing that, continuity of operations to underwrite punishment. Fundamentally, however, the BPR fails to extrapolate beyond broad claims of “detering”. As pointed out by others, the document lacked definitional clarity⁵⁵ – a key requirement for effective communication of deterrence. It failed to define deterrence, did not communicate specific capabilities that would deter, and used the word deterrence against events that are not initiated by humans (bioincidents) and are therefore *undeterrable*. Attempts to increase deterrence capability through the creation of the biodefense council was welcomed by some; though others have argued that the systematic changes multiplied already ongoing efforts, failed to account for others, and further bureaucratized an area that must be able to navigate opaque scenarios decisively.

On newly emerging technologies, policy swung fast. Biden’s AI Executive Order 14110 (Oct. 30, 2023) seeded bio-safeguards⁵⁶ like AI model red-teaming and nucleic-acid screening⁵⁷; DHHS issued updated gene-/genome-synthesis screening guidance (Oct. 2023), and the Office of Science and Technology Policy’s 2024 Framework pushed verifiable screening that major funders (e.g., NIH) began requiring.⁵⁸ In January 2025, the new administration revoked EO 14110,⁵⁹ but in May 2025 issued a separate order tightening oversight and pausing certain dangerous biological research pending new rules—policy churn that nevertheless keeps the focus on blocking pathways to capability.⁶⁰ On the punishment track, Congress advanced the bipartisan BIOSECURE Act to cut federal contracting with designated PRC-linked biotech firms (e.g., BGI, WuXi),⁶¹ and committees kept revisiting and refining the bill into 2025;⁶² DOJ cases (such as June 2025 charges over smuggling the crop-blighting fungus *Fusarium graminearum*) telegraphed legal and economic

⁵⁴ BPR, p.11

⁵⁵ <https://warontherocks.com/2023/09/the-biodefense-posture-review-needs-focus-to-succeed/>

⁵⁶ <https://www.federalregister.gov/documents/2023/11/01/2023-24283/safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence>

⁵⁷ https://bidenwhitehouse.archives.gov/wp-content/uploads/2024/04/Nucleic-Acid_Synthesis_Screening_Framework.pdf

⁵⁸ <https://grants.nih.gov/grants/guide/notice-files/NOT-OD-25-012.html>

⁵⁹ <https://www.whitehouse.gov/presidential-actions/2025/01/removing-barriers-to-american-leadership-in-artificial-intelligence>

⁶⁰ <https://www.whitehouse.gov/presidential-actions/2025/05/improving-the-safety-and-security-of-biological-research>

⁶¹ <https://www.congress.gov/bill/118th-congress/house-bill/7085>

⁶² <https://www.arnoldporter.com/en/perspectives/advisories/2025/08/congress-revisits-the-biosecure-act>

consequences for illicit activity.⁶³ Whether economic consequences truly represent “unacceptable punishment” is debatable, especially considering the ease with which regimes have circumvented such sanctions. Nonetheless, it represents an attempt at “integrated deterrence” that seeks to utilize the U.S.’s broader arsenal in favor of nuclear weapons (or kinetic attacks). These forms of deterrence also allow for a build-up of responses (controlled escalation), which can help communicate credible cost imposition to a wider range of attacks.

The current net result is a posture that preserves the possibility — under Biden, still only in “extreme circumstances”⁶⁴ — that nuclear forces deter strategic-level BW use. However, day-to-day deterrence leans harder on our capability to deny the adversary both the resources and ability to damage the U.S. through resilient operations under CBW conditions, rapid attribution, synthesis-screening rules, research-governance guardrails, and targeted economic-legal tools. Taken together, the U.S. deterrence architecture is eclectic. It pairs declaratory ambiguity at the high end with rule-making, screening, and enforcement at the low end. Its effectiveness hinges less on any single threat than on the visible functioning of the broader system (attribution that stands up in court and in public, enforcement that bites, and allies that adopt comparable screens and standards). Conversely, episodic reversals or gaps between federal guidance and implementation, as well as disconnects between frontier technologies and military response frameworks, invite probing by states and nonstate actors alike. In that sense, the United States ends this period with a posture that is adaptable and usable, but only as durable as our willingness to keep tightening the screws where science, law, and strategy meet.

Some would argue that this architecture has been altogether successful. After all, there have been no BW attacks on CONUS by a state actor since the early 20th century, and BW use by non-state actors has been thwarted (or at least limited) since concerted efforts to deny attacking capabilities were made. In this sense, biological weapons might be seen as the “dog that did not bite”.⁶⁵ However, as previously explained, deterrence is not a state of being but rather a characteristic of a particular practice. This practice must keep up with a changing threat landscape and attempt to preempt rather than to react to emerging capabilities. In the face of 21st-century biotechnological advancements, it is yet to be said that the deterrence for BW has been foolproof. If anything, the last decade has highlighted weaknesses in our theoretical deterrence by denial frameworks and proven that our defenses may not be as impenetrable as believed; an opening that a malicious actor might well seek to exploit.

The COVID-19 pandemic showed the world that the United States’ ability to deny the effects of a pathogen was limited. By late March–April 2020, hospitals nationwide reported

⁶³

<https://www.justice.gov/usao-edmi/pr/chinese-nationals-charged-conspiracy-and-smuggling-dangerous-biological-pathogen-us>

⁶⁴ We await the current administration’s NPR with great anticipation.

⁶⁵ Countering WMDs, p.13

“widespread shortages” of N95s, gowns, and face shields; forcing extended use, reuse, and even decontamination of single-use respirators under emergency guidance (measures the Center for Disease Control (CDC) labeled “crisis capacity”).^{66, 67} The just-in-time, import-reliant supply chain of the Strategic National Stockpile “was not positioned to respond effectively,” concluded one scathing report by the Office of the Inspector General.⁶⁸ DHHS and CDC data show ICU occupancy repeatedly crossed thresholds associated with “strained” conditions (>80%), with nearly half of U.S. acute-care hospitals reaching critical ICU capacity for at least two weeks in the first pandemic winter.⁶⁹ In turn, higher ICU load correlated with excess deaths (75% ICU load resulted in 12,000 excess deaths two weeks later), meaning that even a pathogen with a “low” mortality rate could kill beyond its means.⁷⁰ While programs like Operation Warp Speed were vital in the creation of vaccines, these only became available 11 months after the first infection in the United States. It would take 18 months for fewer than 70% of the population to receive one of the two vaccinations. Suspicion amongst the general public further polarized their adoption. The pandemic cost the U.S. economy a staggering \$16 trillion dollars, according to an estimate from Harvard University.⁷¹ More importantly, it resulted in an excess of 110 million infections and 1 million deaths.

These vulnerabilities notwithstanding, still others have argued that even with the opportunity to do so, a BW attack is simply not an attractive tool because its use is not prestige-endowing.⁷² In this sense, there is no immediate need for a deterrent, because the moral cost (or lack of prestige gain) is in and of itself prohibitive. However, this perspective, too, merits caution. There are non-state actors who do not have the same audience costs as others,^{73, 74} and the biological domain has increasingly become a frontier that states are competing for dominance in. China has pushed biotechnology to the forefront, having established it as one of its “pillars” for the last two 5 Year Plans. In the 2023 BPR, the United States explicitly acknowledges China as a “pacing threat”. Russian leadership, including President Vladimir Putin, has publicly discussed the potential of new “genetic” and unconventional weapons in the future of warfare.⁷⁵ The idea that BWs are a weapon of a dirtier past does not comport with the energy invested by states today in attempting to control their advancement and predict their capabilities before they emerge.

⁶⁶

<https://oig.hhs.gov/reports/all/2020/hospital-experiences-responding-to-the-covid-19-pandemic-results-of-a-national-pulse-survey-march-23-27-2020/>

⁶⁷ https://archive.cdc.gov/www_cdc.gov/coronavirus/2019-ncov/hcp/respirators-strategy/

⁶⁸ <https://www.oversight.gov/sites/default/files/documents/reports/2023-11/42002028.pdf>

⁶⁹ <https://www.medrxiv.org/content/10.1101/2020.12.16.20248366v1.full-text>

⁷⁰ <https://www.cdc.gov/mmwr/volumes/70/wr/mm7046a5.htm>

⁷¹ <https://pmc.ncbi.nlm.nih.gov/articles/PMC7604733/>

⁷² <https://www.nti.org/analysis/articles/disincentivizing-bioweapons-theory-and-policy-approaches/>

⁷³

<https://www.ojp.gov/ncjrs/virtual-library/abstracts/audience-and-message-assessing-terrorist-wmd-potential>

⁷⁴ <https://pubmed.ncbi.nlm.nih.gov/10458956/>

⁷⁵ <https://pubmed.ncbi.nlm.nih.gov/22673989/>

In sum, the historical record yields no single doctrine so much as a toolbox consisting of intermittent nuclear ambiguity, episodic cost-imposition, and expanding denial measures that still falter under attribution and surge-capacity stress. The COVID-19 pandemic underscored how thin this day-to-day denial capability can be, and as biotechnology promises advanced capabilities far removed from the anthrax bombings of the Cold War, we must contend with new styles of attacks and new ways to deter them.

Deterring BWs Today

Both an overt battlefield scenario involving BWs (especially if the opponent is a nuclear state), and covert use (small-scale) lack a credible deterrent response. As emerging biotechnologies lower barriers to entry and blur the line between offense and defense, the foundational principles that once governed deterrence of BWs are coming under acute strain. This section examines how advanced BWs, particularly those enabled by AI, challenge traditional deterrence and render our current models increasingly obsolete. Unless this gap is confronted directly, tomorrow's crises may unfold in a strategic environment unmoored from the assumptions that once kept WMD use at bay.

	Traditional deterrence of BW (Past Assumptions)	Disruptions to traditional deterrence by new technologies
Capability	Punishment credible against large-scale, state BW use.	Small teams/NSAs can cause mass effect, rendering punishment harder.
	Denial worked against known, slowly changing pathogens.	Engineered pathogens can evade current denial techniques.
	C3 assumed intact to coordinate responses.	BW can uniquely target C3 and degrade decision making.
Commitment	Red lines feasible against a small number of state programs.	Many attack vectors make red lines harder to define and signal.
	Past conventional retaliation to CW-use has not changed behavior significantly.	Massive retaliation is not credible across a range of attacks, and limited retaliation continues to not change aggressor behavior.
	Nuclear threat backstopped ambiguity.	Attribution uncertainty invites restraint and makes nuclear retaliation unattractive.
Communication	Limited dual-use scope eased oversight and control.	Massive increase in DURC makes effective control regimes difficult.
	Ambiguous declaratory policies sufficed.	Ambiguous declaratory policies become less credible.
	Denial capabilities seen as broadly robust.	COVID-19 provided proof of weaknesses in denial architecture aggressors could exploit.

A credible deterrent, as explained by Schelling, requires three things: capability, commitment, and communication. Advanced BW challenges each element: capability is undermined when attribution is slow or uncertain, and command, control, and communication (C3) becomes degradable in novel ways; commitment falters when thresholds for attacks are ambiguous and effects unfold stochastically; communication is

complicated by the deep dual-use nature of modern biotech (making bright-line red lines difficult to specify and enforce). Finally, as different actors seek to employ BWs in different ways, monolithic models of deterrence break down and require ever more complex responses, breaking down a state's ability to credibly communicate consequences.

Capability: The limits of retaliatory power in the biological domain

The capability to respond relies upon an agreement on how to respond. If deterrence relies upon unacceptable punishment, one must have an understanding of what constitutes unacceptable punishment for an actor willing to employ biological weapons against the United States. As explained by the Royal United Services Institute, “judgements on rationality, desire, outcomes, and culture are equally essential in assessing which exact tools will change the balance of risk in the adversary's mind.” While the U.S. BPR does outline a process of “integrated deterrence” (ergo, non-nuclear), it is difficult to imagine that an actor willing to use BWs on U.S. soil can be deterred through non-catastrophic means. The ability for actors to accept imposed costs by the United States has been a historical fact. Syria's use of chemical weapons against its own population was met with large conventional attacks,^{76, 77} sanctions,⁷⁸ and political exclusion by the United States.⁷⁹ These efforts raised costs, but did not change behavior. Procurement of chemical weapons and their use against civilians continued.⁸⁰ In fact, it might be telling that Iraq's desire to use BWs was stopped by nothing short of veiled nuclear threats by the United States.

Still, states today have both the capability and legal framework to respond to a large-scale BW attack with overwhelming force, including nuclear weapons. As shown above, the United States has maintained a flexible response posture regarding biological or chemical attacks, leaving the option open to respond with nuclear means in extreme cases; or punish through the more ambiguous mechanism of integrated deterrence.⁸¹ The United Kingdom's 2021 Integrated Review reserves the right to use nuclear weapons against any state that uses

⁷⁶

<https://www.defense.gov/News/News-Stories/Article/article/1144601/trump-orders-missile-attack-in-retaliation-for-syrian-chemical-strikes/>

⁷⁷

<https://www.defense.gov/News/News-Stories/Article/Article/1493718/pentagon-officials-describe-syria-strikes-hope-assad-gets-message/>

⁷⁸

<https://www.federalregister.gov/documents/2013/09/10/2013-22032/determinations-regarding-use-of-chemical-weapons-in-syria-under-the-chemical-and-biological-weapons>

⁷⁹

<https://www.opcw.org/media-centre/news/2021/04/conference-states-parties-adopts-decision-suspend-certain-rights-and>

⁸⁰

<https://www.armscontrol.org/blog/2020-04-13/opcw-investigation-confirms-syria-responsible-three-chemical-attacks-2017>

⁸¹

<https://europeanleadershipnetwork.org/commentary/deterrence-without-destruction-rethinking-responses-to-biological-threats/#:~:text=susceptible%20to%20reverting%20to%20familiar.make%5D%20it>

biological weapons against it, despite otherwise pledging no nuclear use against non-nuclear states.⁸² Russia has signaled an even lower threshold for nuclear retaliation if it perceives an existential threat to itself⁸³ – a category into which BWs could arguably fit. By contrast, China’s declared no-first-use nuclear policy means Beijing professes it would not respond to a biological attack with nuclear weapons (especially if the attacker is a non-nuclear state). Instead, a massive conventional retaliation could still serve the purpose of punishing a BW aggressor.

On paper, then, nuclear-armed states possess an immense punitive capability intended to deter any would-be biological aggressor. However, this theoretical capability rests on unstable prerequisites. The foremost challenge blocking this capability is attribution. Deterrence by punishment only works if a state can confidently identify the attacker and do so swiftly. A nuclear attack leaves a return address; a genetically engineered virus released covertly does not.⁸⁴ If a biological attack’s origin is ambiguous, and a clear establishment of the aggressor occurs on the scale of months or even years, any retaliation (nuclear or conventional) lacks a target. A 2024 workshop conducted at Lawrence Livermore’s Center for Global Security Research underscored this problem: while participants noted that nuclear weapons could in theory deter biological attacks as part of an “integrated deterrence” approach, they also emphasized that “detering or dissuading in a biological threat context without attribution” is highly problematic.⁸⁵ Attribution remains a fraught process, involving complex scientific and investigative processes to pin down responsibility.⁸⁶

Beyond attribution, timing and proportionality further complicate the use of a retaliatory ability. By the time an adversary has been identified, the window for a meaningful counterstrike might have already closed – the attacker’s limited goals already achieved, or the conflict itself resolved. While decision makers are trained not to retaliate meaningfully on fragmentary evidence, it is worth noting that under battlefield conditions, these trainings might fail; these decision makers might face pressure to retaliate on fragmentary evidence, raising the risk of accidental strikes and unintended escalation. History offers sobering proof of this dynamic. In 1994, a plague outbreak in western India sparked rumors of a Pakistani biological attack. Although the outbreak was natural in origin, this misattribution sparked panic and suspicion⁸⁷; a 1996 dengue-fever outbreak in India was then framed by Dr Kamal Datta, director of the Indian National Institute of Communicable Diseases, as a

⁸² European Leadership Network

⁸³

<https://www.armscontrol.org/act/2024-12/news/russia-revises-nuclear-use-doctrine#:~:text=The%20revision%20policy%20also%20states.%E2%80%9D>

⁸⁴ At least not in the same manner where it is immediately identifiable.

⁸⁵

<https://cgsr.llnl.gov/sites/cgsr/files/2024-08/2024-03-Biosecurity-Amidst-Technological-and-Geopolitical-Dynamism-Annotated-Bibliography.pdf>, p.9

⁸⁶ Ibid.

⁸⁷ <https://pmc.ncbi.nlm.nih.gov/articles/PMC1121283/>

“suspicious” outbreak. In a crisis, misidentifying a disease outbreak as deliberate sabotage could trigger an erroneous military response. Recognizing this danger, analysts have urged nuclear powers to “reconsider doctrines that leave open the possibility of a nuclear attack in response to a perceived biological warfare attack”, given the high risks of unclear or incorrect attribution.⁸⁸

Schelling argues that ceding control over the launch of one’s nuclear weapons increases credibility by leaving something “to chance”. Critically, this is only the case when an attacker or an attack is identifiable – usually this means system degradation, or some form of kinetic signature. But biological weapons target neither of these. They target plants, animals, and most importantly, humans.

As a result, a country could lose critical C3 before it even recognizes an attack, preparing the battlefield to the attacker’s advantage. This kind of silent first strike aimed at the mind rather than material is uniquely suited to biology. No other class of weapon can so selectively target the living components of command systems while leaving physical assets intact. Cyber attacks and EMP blasts come close to these effects, leaving the surrounding infrastructure intact while damaging capabilities. However, EMPs cannot distinguish between friend or foe and are a conspicuous event. A bioweapon could theoretically be introduced into only the enemy’s environment (for example, a pathogen that the attacker’s population is vaccinated against, but the defender’s is not)⁸⁹, and unlike cyber attacks, which target software and networks, bioweapons directly target people by covert means (e.g. an infected intermediary or aerosol dissemination at a meeting).

Although these means of attack sound futuristic, Chinese military thinkers have already written about biological warfare in tandem with “cognitive domain” operations, envisioning the use of biological means to weaken an opponent’s decision-making capacity and “achieve new battlefield conditions favorable” to China.⁹⁰ Research into tailored neurotoxic substances or pathogens that target brain function is ongoing in multiple countries, and a paper by the U.S. National Defense University (NDU) lays out a vast arsenal of options that can be used in the cognitive domain, ranging from neuropharmacological agents (drugs), neurotoxins, and neurotechnological systems (lasers, nano-machines).⁹¹ Even temporary

⁸⁸

<https://www.globalasia.org/v16no2/cover/biological-weapons-and-nuclear-deterrence-in-the-pandemic-era-richard-pilch-miles-pomper>

⁸⁹ Such mass vaccination programs would surely become the subject of interest to intelligence services looking to preempt any BW attack. This in turn might allow defending countries to engage in “left of aachoo” strikes; though once again, these preemptive strikes come with their own host of escalation risks.

⁹⁰

<https://www.govinfo.gov/content/pkg/GOVPUB-D109-PURL-gpo194931/pdf/GOVPUB-D109-PURL-gpo194931.pdf>, p,17

⁹¹

<https://ndupress.ndu.edu/Media/News/News-Article-View/Article/2053388/redefining-neuroweapons-emerging-capabilities-in-neuroscience-and-neurotechnolo/>

incapacitation of leaders through a non-lethal bioagent that causes confusion, extreme drowsiness, or disorientation could paralyze the chain of command at a critical moment.

These dynamics carry stark implications for the deterrence of BWs. During the Cold War, both superpowers took steps to ensure that their nuclear forces could not be knocked out in a single blow; this resilience was the cornerstone of deterrence. If one or both sides fear that an opponent might possess a subtle capability to assassinate leadership or disrupt the chain of command biologically, it injects a high degree of uncertainty into this deterrence equation. This could, in turn, pressure decision makers to predelegate launch authority or authorize launch sequences, raising the risk of accidental or unauthorized use of force. Moreover, an adversary might incorrectly perceive a disease outbreak or a sudden sickness in leadership as evidence of an enemy's bioweapon use and rush into hasty action (a variant of the misinterpretation dangers discussed earlier).

Thus, BWs threaten the control dimension of deterrence by targeting our dependence on human decision makers. In the face of growing advancements in AI and the fear of delegated launch authority, military practitioners have emphasized the importance of keeping a human in the loop.⁹² Ironically, this precaution becomes a vulnerability in and of itself as humans become a point of failure. Anxieties about autonomous systems versus human control could be rendered moot if, at a critical juncture, humans themselves are rendered ineffective. The prospect that BWs could force a quiet fait accompli by nullifying an opponent's retaliatory capacity before they even fire a shot is fundamentally corrosive to strategic stability. It incentivizes preemption, delegation, and hair-trigger postures – none of which are desirable.

Finally, even if the capability is intended to deny the effects of an attack rather than to punish them, there are hard ceilings to their efficacy in the BW realm. As explained above, the COVID-19 pandemic underscored that physical protections and countermeasures took months to implement on a national scale. While there have been discussions of “threat agnostic” monitoring programs to preempt attacks before they begin to disperse, daily metagenomic monitoring of bioaerosols remains cost-prohibitive for routine use. The Pentagon Force Protection Agency deemed it “not feasible for daily operations”,⁹³ while national reviews still treat broad genomic biosurveillance as emergent rather than turnkey. Universal vaccines (such as those proposed for all coronaviruses) remain elusive,⁹⁴ and policy suggestions of “immune boosting”⁹⁵ carry risks for fighting-age individuals due to over-zealous immune responses.⁹⁶ At the end of the day, deterrence by denial is a cat-and-mouse game. Defenders publish assays, screens, and detection rules; adversaries

⁹² <https://hcss.nl/news/paul-van-hooft-ai-and-nuclear-weapons-keeping-the-human-in-the-loop/>

⁹³ <https://pubmed.ncbi.nlm.nih.gov/38625036/>

⁹⁴ <https://www.nature.com/articles/s41423-023-01116-8>

⁹⁵ <https://www.darpa.mil/news/2018/bodys-defenses-public-health-threats>

⁹⁶ <https://pmc.ncbi.nlm.nih.gov/articles/PMC4541073/>

adapt with sequence variants and evasion strategies. This adversarial dynamic will now only be accelerated by AI-enabled biological design.

Commitment: Will we really do it?

Even if powerful retaliation options exist, deterrence fails if the adversary doubts one's will to use them. In classic deterrence theory, credibility of the threat is as vital as the capability behind it. Here, too, biological weapons pose a daunting challenge. Would a rational national leader truly order a nuclear strike or massive conventional strike in response to a shadowy bioweapon attack that might kill thousands? The adversary will calculate that in many plausible bio-attack scenarios, the answer is no, or at least not a certain yes.

Several factors erode the ability to commit credibly. First is the sheer diversity of possible biological attacks⁹⁷. Unlike the binary cataclysm of a nuclear strike, bioweapons occupy a spectrum from minor to apocalyptic. At one extreme, one could imagine a large-scale attack causing a deadly pandemic, which clearly represents a grave strategic threat. At the other extreme, there are ethnically targeted bioweapons, crop-destroying pathogens, and location-specific attacks (such as the targeting of key workers at a munitions depot). These represent attacks in the grey-zone; that is to say, gradualist campaigns by state and non-state actors that combine non-military and quasi-military tools that fall below the threshold of armed conflict, aimed at thwarting, destabilizing, weakening, or attacking an adversary, often tailored toward the vulnerabilities of the target state.⁹⁷ BW greyzone warfare could include covert operations to ruin a single city's water supply, or a bespoke pathogen that blights a nation's staple crop for a season, or a non-lethal agent that temporarily incapacitates the occupants of one airbase. Such actions could have significant cumulative effects yet fall short of overt mass casualty strikes. In such cases, it becomes very difficult to create a 'red line' over such a vast range. If national policy only promises devastating retaliation for massive, Cold War–style biological attacks, an adversary may reason that smaller-scale bio-aggressions will fall below the threshold and escape punitive response. Indeed, analysts warn that states may find “bioweapons useful for tactical purposes — for example, to carry out a covert, plausibly deniable economic attack” or other limited goals.⁹⁸ Such gambits exploit the credibility gap: will a democracy incinerate a city in return for crop blight or a mysterious epidemic that sickened a few hundred people? Will the aggressor relent due to the fear of conventional strikes, sanctions, and international opprobrium?

97

<https://www.atlanticcouncil.org/blogs/new-atlanticist/todays-wars-are-fought-in-the-gray-zone-heres-everything-you-need-to-know-about-it#:~:text=and%20human%20rights?-.What%20exactly%20is%20a%20%E2%80%9Cgray%20zone%2C%E2%80%9D%20and%20how%20has.its%20allies'%20ability%20to%20respond.>

98

https://www.nti.org/wp-content/uploads/2024/12/Disincentivizing-Bioweapons_NTI-Book_12.10.24-Final.pdf, p.15

Conversely, if a state tries to widen its declared red line and signals that any use of bioweapons, however small, will trigger massive retaliation, that threat will also lack credibility. Enforcing an absolute red line would require an almost unimaginable level of resolve (or recklessness), essentially committing to (possibly) nuclear escalation over even a relatively contained incident. Adversaries may simply not believe this commitment, especially if the deterring state is constrained by moral, legal, and alliance considerations. As Eve Siegmann notes, expanding nuclear deterrence to cover biological threats risks eroding the nuclear taboo and would strain credulity, given the global norm against nuclear use.⁹⁹ This stance, while normatively sound, signals to adversaries that nuclear retaliation is a highly unlikely response to anything but the most cataclysmic bioweapon attack. Extending this protection to allies becomes even more contentious. In a scenario where an ally might be attacked by BWs, the U.S. would face all the same attribution and communication difficulties that such an attack on its own soil would bring, but compounded by unknown escalation risks. For example, in a scenario where the DPRK covertly uses BWs in an attack on South Korea, Washington would have to weigh whether retaliation might trigger North Korean strategic escalation.

An all-encompassing red line also suffers from the attribution puzzle. Given the current difficulties in tracking the origins of pathogens and ascertaining whether their presence is natural or not, a rational adversary would be tempted to act if they think they can hide their hand long enough to achieve victory. Real-world cases already prove this incentive. The U.S. has publicly accused nations like Russia, North Korea, and Iran of maintaining clandestine bioweapons programs despite the Biological Weapons Convention ban.¹⁰⁰ Those regimes presumably calculate that they can avoid decisive proof or consequences, thus undermining the credibility of international deterrence efforts. A 2023 NTI study on shaping state intentions reports that for deterrence to work, actors must believe they will “likely be caught and face consequences” for pursuing bioweapons.¹⁰¹ The world’s verification and forensic capabilities for biothreats, while improving, are still not infallible. A sophisticated state attacker might employ cut-outs, false flags, or simply exploit the chaos of a disease outbreak to mask their involvement. If leaders in Pyongyang or elsewhere are confident that any retaliation will either miss the true culprit or come too late, the deterrent loses its bite.

To shore up credibility under these conditions, a deterring state has two unpalatable options. One is to be extremely explicit in promised retaliation, essentially threatening to respond immediately and overwhelmingly to even a whiff of a BW attack [99]. This “shoot first, ask later” posture would involve pre-delegated authority or standing plans to strike on suspicion, raising the risk of horrific mistakes. Needless to say, few democracies or

99

<https://europeanleadershipnetwork.org/commentary/deterrence-without-destruction-rethinking-responses-to-biological-threats/>

100

<https://www.state.gov/adherence-to-and-compliance-with-arms-control-nonproliferation-and-disarmament-agreements-and-commitments/>

¹⁰¹ NTI report, p.64

responsible powers can commit to this without undermining their own values and the trust of their publics and allies. The second option is to invest in far better attribution and verification capabilities, so that the response can be both precise and justified. If effective, such mechanisms could buttress credibility by assuring any perpetrator that they will be identified in short order. Similarly, advances in pathogen forensics (the genetic “fingerprinting” of strains) aim to narrow down sources of an outbreak to a particular lab or region. However, until such capabilities are proven and widely accepted, they may not weigh heavily in an adversary’s calculations.

In sum, the credibility of retaliation against bioweapons use is inherently fragile. Adversaries can exploit the gray zones using methods whose impacts are significant but not unmistakably attributable or strategically obvious, betting that the victim will be paralyzed by uncertainty or restraint. As Thomas Schelling observed, deterrence often hinges on a “threat that leaves something to chance,” injecting risk to persuade the adversary. With biological attacks, however, so much is already left to chance (chance of discovery, chance of public attribution, chance of escalation) that the defender’s threats ring hollow. This credibility gap is a core reason why traditional deterrence by punishment alone is inadequate for tomorrow’s bioweapon threats^[OBJ].

Communication: Red Lines and Miscalculation

As discussed above, states attempting to deter a BW attack will struggle with creating red lines that are either too wide or too narrow. Too broad, and a state risks creating a tripwire whereby an accident or misperception could have catastrophic consequences. A novel disease outbreak, an unusual cluster of crop failures (or individuals found with the required pathogens to induce it)¹⁰², or a natural pandemic could all be mistaken for a bioweapon attack under a hair-trigger doctrine. In an extreme case, a naturally occurring pandemic might be misread as deliberate and provoke a military overreaction. Scholars have warned that in an age of frequent pandemics, countries must be “vigilant in determining the precise cause” of outbreaks and restrained in responding to perceived biological attacks with nuclear escalation (or any kinetic response).¹⁰³ On the flip side, overly narrow or caveated red lines are also problematic. If a state declares it will retaliate only for a catastrophic biological attack causing mass casualties, this might be interpreted by adversaries as permission to undertake smaller attacks. Any threshold leaves room for an attacker to calibrate just below it. The Soviet Union, for example, was accused of using trichothecene “yellow rain” toxins in Southeast Asia in the early 1980s – a form of chemical-biological attack that was deniable and not severe enough to prompt U.S. retaliation, yet effective in terrorizing insurgent populations. The debate as to whether or not the Soviet Union truly

¹⁰²

<https://www.justice.gov/usao-edmi/pr/chinese-nationals-charged-conspiracy-and-smuggling-dangerous-biological-pathogen-us>

¹⁰³ Global Asia piece

did employ trichothecene is still ongoing to this date.¹⁰⁴ Similarly, if a state like Russia considered using a non-contagious but lethal agent against a single high-value target (say, contaminating an opposition leader or a military base), it might gamble that this would not clearly violate a vaguely worded Western red line. The range of possible BW scenarios is so large that any finite list of prohibited actions will be incomplete. This places an extra burden on communication: deterrence messages must convey intent and resolve across a continuum of threat, without being either blindly universal or too case-specific.

Compounding these issues is the dual-use nature of biological research, which further muddies the signaling of intentions. In the nuclear domain, certain actions (like testing a nuclear device or moving missiles) send clear signals. In biology, the same research that can produce a weapon can also be purely defensive or civilian in purpose. If a nation publicly states a “red line” against developing certain pathogens, how does it then treat legitimate biodefense research dealing with these pathogens? For instance, banning all dual-use research of concern (DURC) might seem like a way to signal zero tolerance for bioweapons work. But as a group of researchers from the National Academy of Sciences argued in a recent report, “biotechnology [...] is a dual-use technology”.¹⁰⁵ If responsible states curtail their biodefense programs for fear of crossing a perceived red line, they may handicap themselves in preparedness, all while rogue actors continue covertly. Meanwhile, such a prohibition would not only be unverifiable, it would drive weapons-related work deeper underground, making it even harder to monitor threats that states would then be less prepared to face.

If states do want to communicate readiness, they must be careful to avoid a perverse arms race in signaling capability. Effective deterrence by denial – i.e., convincing adversaries that any BW attack will fail – requires showcasing robust defenses such as rapid detection systems and stockpiles of medical countermeasures.¹⁰⁶ While this approach focuses on resilience rather than threatening punishment, communicating it still has complex effects on adversaries. Publicly advertising advanced biodefense capabilities can look like advertising latent bioweapon capacity. The line between defensive and offensive know-how is thin or nonexistent.

Demonstrating an ability to rapidly produce vaccines for any pathogen implies a sophisticated biotechnology apparatus that could potentially create novel pathogens as well. It also implies an understanding of that pathogen and methods of its production sufficient to use it as a weapon. This can breed mistrust: other states may view robust U.S. or allied biodefense programs as cover for their own bioweapons, potentially spurring them to accelerate their efforts. In deterrence theory, such action-reaction cycles are reminiscent of missile defense plans and their destabilizing effect. Biological arms racing could be an unintended byproduct of loudly proclaimed biodefense prowess.

¹⁰⁴ <https://www.nonproliferation.org/wp-content/uploads/npr/81tucker.pdf>

¹⁰⁵ National Academy paper, p.xx

¹⁰⁶ <https://warontherocks.com/2021/09/a-deterrence-by-denial-strategy-for-addressing-biological-weapons>

A further complication is the temptation to wield dual-use research of concern (DURC) as a performative deterrent to subtly signal offensive latency by showcasing cutting-edge synthetic biology, rapid pathogen engineering, or AI-enabled design pipelines.¹⁰⁷ The theory is that such “translucent” capability signaling will dissuade aggression without ever requiring use. In practice, this gambit is unstable. First, DURC display does enable attribution, but only up to a point: if a uniquely engineered feature appears in a future pathogen, observers can plausibly infer state responsibility or at least state tolerance/involvement. Yet if the deterrent works, no use occurs and attribution never matures into proof; the same translucency that is meant to deter also preserves deniability if something does happen, because DURC outputs are widely reproducible across the global bioeconomy. Second, devastating, immediately lethal biocapabilities, the only ones likely to mimic nuclear “assured destruction” signaling, cannot be credibly “flexed” without egregious norm violations or catastrophic proliferation externalities; merely being ‘advanced at biotech’ neither singles a culprit *ex post* nor creates the crisp, audience-convincing threat that nuclear forces do. The result is a deterrent that is strategically latent but communicatively ambiguous: it may raise adversaries’ general caution while still failing to answer, definitively, whether a pathogen incident should be attributed to (and punished as) state action.

A final hurdle in a state’s ability to provide clear communication is the increasing diversity of actors and strategies in the biological weapons landscape. Traditional, monolithic deterrence frameworks were largely constructed with symmetric state adversaries in mind, ergo, actors who shared broadly similar incentives, risk calculations, and thresholds. Modern BWs expand the arsenal of options, however. Non-state groups may seek spectacular violence without fearing retribution, small states might leverage BW to offset conventional asymmetries, and great powers may exploit plausible deniability to erode rivals’ political cohesion without triggering direct conflict. Each scenario demands a distinct deterrent posture, and as these strategies multiply, the clarity of any single red line fades. States attempting to communicate deterrence effectively must therefore strike an increasingly complex balance, sending tailored, credible signals to disparate adversaries without diluting the fundamental message of resolve. In practice, this complexity often fragments deterrence, weakening its credibility and opening exploitable gaps through which malicious actors may slip.

In sum, communicating deterrence in the BW arena (be it as attacker or defender) is difficult. Clear messaging is needed to delineate red lines that, when drawn too finely, allow adversaries to exploit what’s left unsaid, and if drawn too broadly, invite miscalculation and dangerous escalation. The inherent dual-use of biotechnology complicates signals of intent, and efforts to deter by denial require delicate, precise framing to avert the destabilizing effect they attempt to address in the first place.

¹⁰⁷ <https://arxiv.org/abs/2505.17154>

Shaping the Battlefield and Inadvertent Escalation

Biological attacks on command structures blur the tactical/strategic divide in conflict. A limited bio-operation aimed at “shaping the battlefield” (to borrow the American term) such as disabling a regional commander or neutralizing the pilots of an air wing, could yield a strategic advantage out of proportion to its scope. This effect is in fact generalizable outside of command structure attacks. Introducing hydrocarbon-eating bacteria into a forward operating base's fuel supplies could ground forces in the region and lose a state's competitive advantage. Because BWs are highly operational, they can create extremely undesirable outcomes for states on the battlefield, potentially tipping a local conflict and thereby escalating to a larger conflict if the pressured state resorts to drastic measures.

History once again provides valuable insight, here. Declassified Warsaw Pact planning materials and post-1989 testimonies suggest the East German Stasi kept Einsatzkommandos that, in the event of a general European war, were tasked with deep operations in NATO rear areas, including sabotage, assassinations, and the possible use of toxic or biological agents to paralyze command-and-control and logistics and strike fear in local populations.¹⁰⁸ The mere fact that such options were canvassed underscores that, even before today's advancements in synthetic biology, planners already conceived of biological means as battlefield-shaping tools, not just strategic terror weapons.

Deterrence theory struggles with such incremental, ambiguous aggression because it is neither wholly in the peace realm nor an obvious *casus belli*. It creates a grey-zone conflict where the defender's retaliation options are unclear. If they respond conventionally, they might not undo the damage in time; if they respond with strategic force, they might be seen as overreacting or even be unsure at whom to aim. And yet, if a country does nothing, will a precedent have been set for future attacks?

As shown above, classical deterrence frameworks struggle greatly to contain the threats posed by biological weapons. Capability to punish, credibility of will, clarity of communication, and reliable control are each weakened when the weapon in question is a microscopic pathogen rather than a missile. Tomorrow's bioweapons are likely to be stealthy in development, ambiguous in use, and potentially devastating in impact. They can be tailored for covert operations that fall in the grey-zone of conflict, or engineered for mass casualty effects that rival nuclear fallout. This diversity and dual-use nature mean that the deterrence strategies that kept nuclear peace for decades need significant adaptation in the biological domain.

¹⁰⁸ Implied proof exists in the form of CIA documents discussing the use of biochem by the Soviets and their capability to use it “selectively and in the nonnuclear phrase under some circumstances”. <https://www.cia.gov/readingroom/docs/CIA-RDP87T00217R000200070009-1.pdf>, p.4, p.12

Rethinking Deterrence for a New Era

The strategic shift for the U.S. must be to dissuade, rather than deter. Damage of a BW attack cannot be made entirely null, therefore states must raise the cost of BW acquisition so that its limited utility does not outweigh the price of a program. Crucially this must happen *before* an actor decides to use a BW. Current policy suggestions stress the importance of shared governance, international cooperation, and adherence to existing norms. While these arguments are not without merit, their implementation is measured on the scale of years, and we may not have the luxury of time. This section will detail several policy recommendations for U.S. officials in the short and long term that can help protect the United States and its allies from BW attacks against a range of actors while serving as a practical example for other states to follow.

Short Term	Detect BSL-4 construction early	International SME engagement	Map benign-to-malicious pipelines	Data control	Black market surveillance
Medium Term	Focus oversight in high-risk locations	Pre-commit cost imposition	Monitor erosion of tacit knowledge barriers	Establish joint responses and countermeasures	
Long Term	Strengthen control regimes	Build international norms			

Short Term

In the short term, global powers will be the first to access the most advanced BW capabilities through their cutting-edge research in biotechnology and their use of expert capital across a variety of domains. Increasing global awareness of foreign programs will be key to monitoring, tracking, and assessing DURC. To do this, the U.S. should:

- 1. Detect BSL-3/3+/4 constructions early:** Use national technical means¹⁰⁹ combined with established signatures for BSL-3/3+/4 labs (such as high-containment construction with air handling, negative-pressure systems, and specialized waste treatment) to track new facilities in countries of interest. Ensure integration into an international network of biotechnology research laboratories, to quickly spot research that is done outside of these networks. Raise alarm when these constructions are not reported in yearly BWC CBMs, or contain offensive research

¹⁰⁹ National technical means of verification are monitoring techniques such as satellite imagery employed to verify adherence to international treaties.

indicators (such as aerosolization chambers for animal-testing devices to similar ends), or create biosecurity risks. The ability to hold these non-reported facilities at risk will be pivotal in maintaining leverage in future negotiations.

2. **Augment sparse technical collection with SME engagement:** Rather than barring critical research across the world to protect ourselves from the ‘other side’ of DURC, we must encourage and maintain strong scientific collaboration through international engagement and participation in innovative research with both partner and adversarial institutions for relevant information gathering and data sharing.
3. **Preempt “benign-to-malicious” pipelines:** for genuine public-good research with plausible misuse pathways, develop cooperative DURC guidelines that can be implemented in reviewing research. Examples could include funding red-team assessments that identify the minimal additional steps an adversary would need to weaponize it. Make all funding (federal and private) contingent on such reviews can incentivize both academia and industry to meet these goals. Issue quiet alerts to providers and regulators so that when the research goes public, defenders already know what signatures to watch for (especially in smaller states that may apply no-longer-cutting-edge methods opportunistically).
4. **Push for better data control:** Many of the most dangerous consequences of ML rely on access to sensitive biological data. If the data AIs are trained on is restricted to what is already found in the open literature, then their outputs (even if faster) will not be more dangerous than what non-state actors are currently capable of. As of now, there are no international export control regimes that explicitly require the control of immaterial transfers of data. Entities that produce this data have an interest in creating internal systems of control, or agreements between each other, to prevent adversarial use.
5. **Black market vigilance:** Using black markets and secondary markets (such as eBay) to track exporters of key material technologies related to BW production, as well as smugglers of sensitive data that can empower MLs for DURC. The key to such surveillance is not to (outright) shut down buyers, but to identify and disrupt sellers, as well as early identification of dangerous actors. Using this information, one can build databases of individuals and organizations who are working as dangerous biotechnology proliferators (possibly outside of the entity list) and shared internally with partners.

Medium Term

In the medium term, smaller states will likely have a unique interest in BW capability as an asymmetric deterrent. In the medium term, these states will benefit from the dissemination of “cutting-edge” research. Given the capacity and capital available, smaller states can likely emulate but are less likely to lead in innovation in these domains. Danger exists if previous research has been ineffectively red-teamed, and the manner in which it can be used for malicious intent is unclear, leading to longer attribution windows in case of use.

1. **Focus oversight where risk is highest:** Identify states most likely to see advanced BW as an asymmetric hedge and make market access and tech cooperation

contingent on concrete safeguards (e.g., universal sequence screening, verifiable lab-of-origin traceability, external audits). This pairs strategic foresight with enforceable conditions rather than generic watchfulness. Attempting to preemptively secure such labs through cooperation, however, can (and has¹¹⁰), be used by adversarial nations for disinformation tactics. In these cases, engagements between U.S. partners and local populations are key in retaining trust and transparency.

- 2. Establish Joint Responses and Countermeasures:** Currently council resolution 1540 creates a framework for the investigation of alleged BW use, but does not specify actions after use has been established. Using resolution 1540, the U.S. and its partners must create plans of action for how to respond once BW has been established. This framework in turn can bolster the cost imposition strategies of the U.S.
- 3. Pre-Committed Cost Imposition:** Using the U.N. Security Council to define misuse categories (e.g., covert human experimentation; unregistered high-risk constructs; deliberate bypass of screening) and tie them to a menu of escalating, coalition-backed penalties¹¹¹ as defined by council resolution 1540. These should include export denials, funding cut-offs, seizures, targeted sanctions, all the way to kinetic strikes. Announce the categories and the menu now, then apply them through the UNSGM when evidence crosses agreed thresholds. These ‘menus’ can be tailored to countries of concern, as cost imposition strategies will be unique to each actor.
- 4. Monitoring erosion of Tacit Knowledge barriers:** As of now, great powers likely have an intellectual monopoly on the tacit knowledge necessary for advanced BW development. However, as vision models and robotics learn lab procedures, funding red-team work to identify which tacit bottlenecks are dissolving first will be crucial. Where feasible, add firmware/software interlocks to benchtop tools and protocols most likely to “compress” skill barriers. While no permanent barrier is possible, raising the effort required will make covert BW development using ML more difficult and easier to detect. Finally, monitoring “talent flight” of key researchers and preventing intellectual property theft by individuals involved in DURC can help preempt possible new programs ahead of time.

Long Term

In the long term, non-state actors pose the most unpredictable and ideologically driven threat in the BW landscape. Their motivations are less susceptible to traditional deterrence, and over time, they will benefit most from the democratization of dual-use biotechnologies. In the long term, the focus must shift toward denying these actors the opportunity to acquire, produce, or deploy BW capabilities not by attempting to eliminate intent, but by constraining access to enabling tools, knowledge, and sponsorship.¹¹² This requires both norm-shaping among states and anticipatory disruption of emerging access pathways.

¹¹⁰ <https://apnews.com/article/0cf158200e674f41bd3026133e5e043d>

¹¹¹ Using existing export control groups such as AG as a base to form such coalitions would be fruitful.

¹¹² Most importantly by asymmetrically-hedged state actors

- 1. Strengthening Control Regimes:** Strengthening control regimes will always require balancing security with access to vital biological research. Neither unfettered access nor draconic restrictions on key technologies and knowledge will bring about a safer world. Groups of SMEs such as the Australia Group are already leading vital efforts in establishing effective control regimes to stop the spread of dangerous, enabling technologies. These groups must be empowered to provide more than just recommendations to governments; they must be enabled to set specific policy guidelines in collaboration with relevant national security branches. Current patchwork regimes for the screening of DNA synthesis and cloud labs must be unified under a common federal governing organization to streamline oversight, enforce consistent standards, and reduce loopholes that adversaries could exploit. Operating under a shared framework is essential not just for efficiency but to ensure that disparate national or institutional definitions do not create soft targets for malicious actors, or create unnecessary ambiguity.
- 2. Build International Norms:** The cost for NSAs can be uniquely raised by building strong international norms against the development and use of BWs. While they may have incentives to develop or use BWs, NSAs also draw disproportionate international scrutiny for doing so. By strengthening nonproliferation norms, the difficulty in acquisition by proliferating state actors, as well as the cost of use for NSAs, can be drastically increased. To do this, the international community must resist the allure of an arms race when faced with new operational tools. The only effective way to do this is through trust and verification; current mechanisms under the BWC provide some frameworks for how this can be done (CBMs), but these agreements must be given teeth to be substantial. Using other WMD treaties as examples (such as the Treaty on the Prohibition of Nuclear Weapons), the BWC must integrate cost-imposing measures upon non-compliant states. More broadly, the BWC cannot fulfill its mission in its current capacity; it requires institutional reform, stronger enforcement mechanisms, and sustained political commitment to adapt to the age of dual-use AI-driven biology.